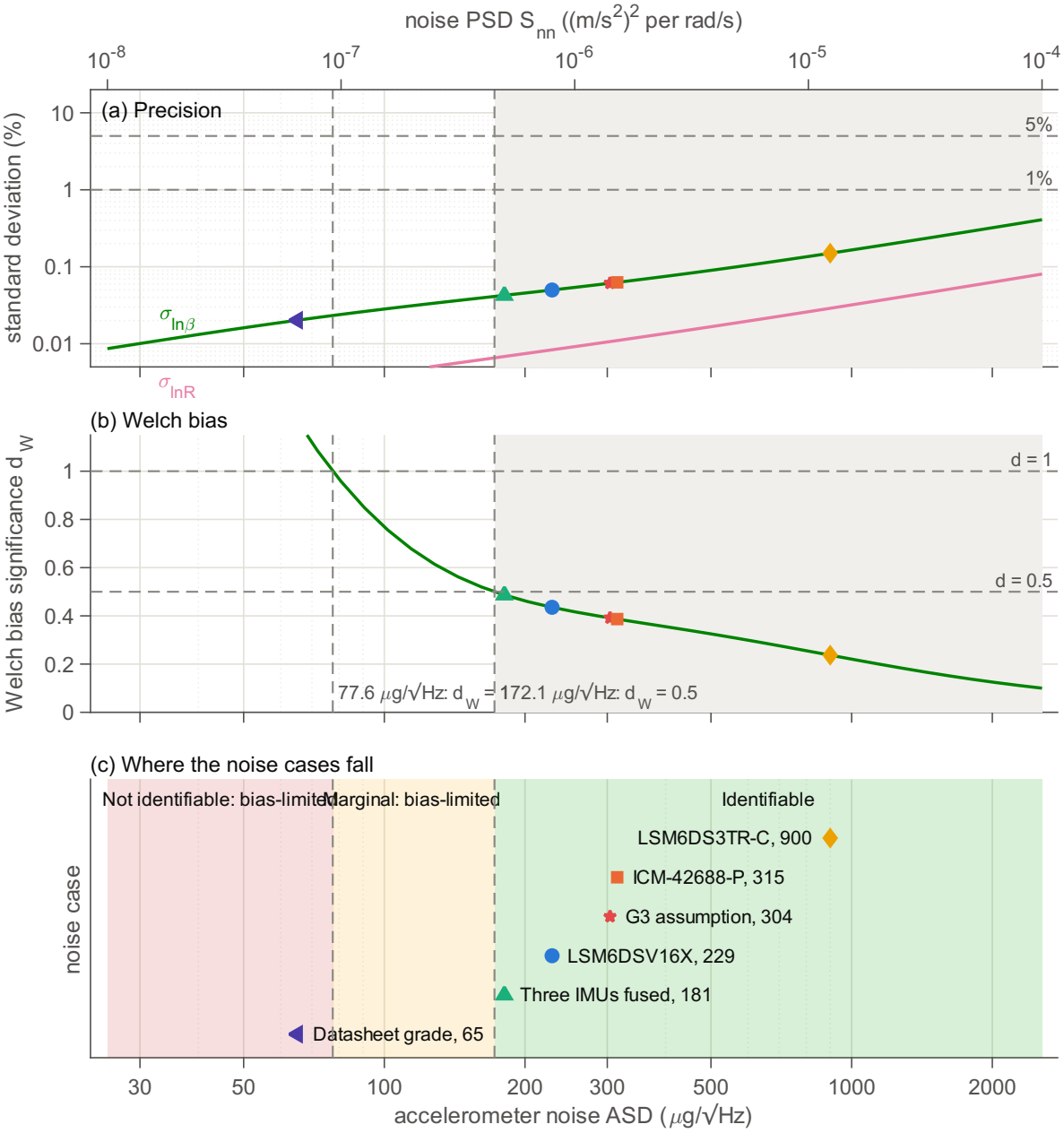

noise PSD S_{nn} ($(m/s^2)^2$ per rad/s)
10^{-8}
10^{-7}
10^{-6}
10^{-5}
10^{-4}
(a) Precision
standard deviation (%)
10
1
0.1
0.01
5%
1%
$\sigma_{\ln\beta}$
$\sigma_{\ln R}$
(b) Welch bias
Welch bias significance d_W
1
0.8
0.6
0.4
0.2
0
d = 1
d = 0.5
77.6 μg/√Hz: d_W = 1
72.1 μg/√Hz: d_W = 0.5
(c) Where the noise cases fall
Not identifiable: bias-limited
Marginal: bias-limited
Identifiable
noise case
LSM6DS3TR-C, 900
ICM-42688-P, 315
G3 assumption, 304
LSM6DSV16X, 229
Three IMUs fused, 181
Datasheet grade, 65
30
50
100
200
300
500
1000
2000
accelerometer noise ASD (μg/√Hz)